

Model Report: CNN-BiLSTM Field-Level (temporal)

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Script: /home/mmann1123/Documents/github/South_Africa_Crop_Comp/deep_learn/src/Deep Learning/Pixel_Field_Level

Data Split:

train_count: 2653
val_count: 664
test_count: 830
random_seed: 42
split_method: fid-wise (field-level training)

Metrics (test set):

Accuracy: 0.7181
Cohen Kappa: 0.6096
F1 Weighted: 0.7284
F1 Macro: 0.6429

Training Time: 53s (0.9 min)

Hyperparameters:

architecture: CNN(1D, 6->64, k=5) + BiLSTM(64)
training_level: field (pixel means per FID)
conv_filters: 64
lstm_hidden: 64
kernel_size: 5
dropout: 0.3
lr: 0.001
optimizer: AdamW (wd=1e-4)
scheduler: CosineAnnealingLR
loss: WeightedFocalLoss(gamma=2.0)
features: raw reflectance (unscaled)
epochs: 100
patience: 15
batch_size: 128
n_models: 5
seeds: [42, 101, 202, 303, 404]

Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Barley	0.605	0.669	0.636	133
Canola	0.878	0.860	0.869	100
Lucerne/Medics	0.921	0.845	0.882	388
Small grain grazing	0.281	0.432	0.340	74
Wheat	0.530	0.452	0.488	135

Confusion Matrix

